Utsab Karan

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Education

Indian Institute of Technology Kharagpur — Kharagpur, India
Integrated Masters in Electronics & Electrical Communication Engineering; GPA: 9.50/10 — *Aug. 2023 – Present*
Specialization in ***Vision and Intelligent Systems***

Experience

Vision and Artificial Intelligence Lab, CDS Department — Indian Institute of Science
Research Intern | Guide: Prof. Venkatesh Babu — *May 2026 – Present*

- Benchmarked SOTA data pruning strategies on **ImageNet** and **CIFAR-100** using **ResNet** architectures
- Working on **continual** and **data efficient learning** methods for vision models, analysing effects on **data scaling**

Autonomous Robots and Multi-robot Systems Lab, SysCon — IIT Bombay
Research Intern | Guide: Prof. Leena Vachhani — *May 2025 – Dec. 2025*

- Simulated crowd environments in **Gazebo/ROS2**, analysing human-robot interaction scenarios with SFM metrics
- Worked on gap-based **topological mapping** with safe kinodynamically feasible planning and control
- Validated exploration and navigation algorithms on **TurtleBot3** hardware, benchmarking against **APFs**

Autonomous Ground Vehicle Research Group — IIT Kharagpur
Vision Head | Guide: Prof. Debashish Chakravarty — *Apr. 2024 – Present*

- Deployed **monocular depth** estimation models and compared outputs with Intel RealSense **stereo depth**
- Pipelined ArUco marker detection, grid mapping and autonomous navigation for **TurtleBot** simulations.
- Developed **Pure Pursuit** and **Stanley** controllers for F1tenth racing simulations
- Tested RL algorithms including **DQN**, **A2C** and **PPO** on OpenAI Gymnasium environments

Publications

1. Schneider, Canzano, Hou, Peng, Deepu, **Karan**, et al. "Visual Robustness and Neural Alignment in a Shared Foraging Task: The Mouse vs. AI Benchmark." Accepted at the **NeurIPS 2026 Evaluations and Datasets Track**. [**Paper**]
2. Moshayedi AJ, Roy AS, **Karan U**, Zhang M, Bassir D. *"An Integrated Beetle Antennae Search–Enabled Navigation Framework for Omnidirectional AGV Mobile Robots in Unknown Environments"*. **EAI Endorsed Transactions on AI and Robotics** 2026. [**Paper**]
3. K. Trivedi, **U. Karan**, A. Sudarshan and L. Vachhani. "Safe Indoor Exploration and Sparse Topological Mapping with Integrated Planning and Control," *11th Indian Control Conference (ICC)*, Bangalore, India, 2025. [**Paper**]

Projects / Competitions

AlphaZero Implementation for the Black-Hole Game [Link] — IIT Kharagpur
The KGPian Game Theory Society — *Jan. 2026 – Feb. 2026*

- Implemented a custom **Gymnasium** environment and trained a Deep Q-Network agent via **self-play** RL
- Extended to AlphaZero-style framework with a **dual-headed** ResNet network and **MCTS-guided** action sampling
- Parallelized self-play games for faster replay buffer accumulation and convergence of the learned policy

Diffusion Language Model from Scratch [Link] — IIT Kharagpur
Self Project — *Dec. 2025 – Present*

- Implemented a continuous **diffusion-based** language model with a **BERT**-style (∼15M parameter) denoiser network
- Implemented end-to-end embedding learning, **posterior regularization** and clamping-based decoding
- Trained on **ROCStories** and **E2E** datasets, analyzing token clusters via t-SNE and optimizing for training stability

Monocular Depth Estimation for Edge Devices [Link] — IIT Kharagpur
Autonomous Ground Vehicle Research Group — *Sept. 2025 – Nov. 2025*

- Implemented a **UNet** style (∼3.5M parameter) depth estimation model with a pretrained **MobileNetv2** encoder
- Designed a light-weight **4-stage** decoder with skip connections, **inverted residuals** and **pixel shuffle** upsampling
- Trained with a joint depth and segmentation based log-space **BerHu** loss on the **KiTTI** and **CityScapes** datasets

Beatle Antenna Search Optimisation for Robotic Navigation [Link] — Oulu University
Remote Collaboration | Prof. Ata Jahangir Moshayedi — *June. 2025 - Aug. 2025*

- Implemented the Beetle Antenna Search(BAS) algorithm for optimising **PID** based controls
- Developed a BAS optimised **A* - path planning** algorithm for obstacle avoidance and safe navigation back to track
- Tested and compared **robustness** of the proposed framework across different environment difficulty configurations

Pure Pursuit Controller — F1Tenth [Link] IIT Kharagpur
Autonomous Ground Vehicle Research Group *Aug. 2024 – Sept. 2024*

- Designed a Pure Pursuit controller for **F1Tenth** racing simulations
- Evaluated performance across multiple tracks using **RViz** and large waypoint datasets
- Integrated ROS libraries, transformations, message formats and node parameters

Coursework

Department Courses: Advanced Calculus, Numerical Analysis, Probability and Statistics, Linear Algebra, Optimization Methods, Analog Circuits, Digital Electronics, Network Theory, Signals and Systems, Systems and Control, Electromagnetic Engineering, Communication-I, Communication-II, Digital Signal Processing - I, Digital Signal Processing - II, Algorithms, RF and Microwaves, VLSI Design, Computer Architecture
Electives: High Performance Scientific Computing, Information Theory and Coding
Others: Game Theory-I (Stanford), Programming for Robotics (ETH Zurich), CS182(UCB): Deep Learning, CS285(UCB): Deep Reinforcement Learning, Flow Matching and Diffusion Models(MIT 6.S184)

Course Projects Summary

Communications Lab: Designed circuits for realizing various signal modulation techniques. **Digital Signal Processing Lab:** Implemented FIR and IIR Filters. Designed a Dual Tone Multifrequency (DTMF) Coder & Decoder in Matlab. **DIY Lab:** Built an Object Tracking Aim Bot using OpenCV and IoT integration. **High Performance Scienftific Computing:** Parallelized Sparse Matrix Solvers using OpenMP, MPI and CUDA. **Embedded Systems Lab:** Designed a real-time clock using Assembly language and implemented on the 8051 test board. **VLSI Design Lab:** Designed and Verified Layouts for Gates and Ring Oscillators.

Technical Skills

Languages: Python, C/C++, CUDA C, Arduino, Verilog, OpenMP, MPI
Libraries: NumPy, Pandas, PyTorch, Matplotlib, OpenAI Gymnasium, D4RL
Frameworks/Software: ROS1/ROS2, Gazebo, RViz, Vivado, LTSpice, Simulink, Docker, Git, Linux
ML/DL: CNNs, Transformers, Diffusion, Generative Modelling, ResNets, RL, Imitation Learning

Awards and Achievements

- **2nd** in Robust Visual Foraging Competition @ **NeurIPS 2025** [Link]
- **5th** in Qualification and **7th** Globally — Roboracer Sim Racing League @ **CDC-TF 2025** [Link]
- **4th** place in Hardware Modeling in **General Championship 2025** [Link]
- **1st** in Qualification and **9th** Globally — Roboracer Sim Racing League @ **ICRA 2025** [Link]
- Ranked **15** among 124k candidates in **WBJEE 2023**
- Ranked **1784** among 1.27M candidates in **JEE Mains 2023**
- Ranked **1596** among 190k candidates in **JEE Advanced 2023**

Extracurriculars

- I am the current **President** of the **KGPian Game Theory Society**. Previously helped organize events like Treasure Hunt and Stratathon which saw participation of 700+ students promoting the art of strategic thinking. I have also headed the Academic Team in their research endeavors in the domain of **game theory** and decision-making.
- Participated in Tech Quiz, Biz Quiz and Basketball in Inter-Hall General Championships. I also led the **Hardware Modeling** Team of my Hall and secured 3rd position(unofficial) in the General Championship 2026.